

MICROTUBULE DYNAMIC INSTABILITY AND GTP HYDROLYSIS

Harold P. Erickson and E. Timothy O'Brien

Department of Cell Biology, Duke University Medical Center, Durham,
North Carolina 27710

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PERSPECTIVE AND OVERVIEW

Over the past two decades, our understanding of microtubule dynamics has progressed through three models, each postulating an increasing level of activity. First, in the early 1970s researchers demonstrated that microtubules could be reversibly assembled from soluble tubulin subunits, but at steady state the dynamics were thought to be limited to the an equilibrium exchange of subunits between the soluble pool and the microtubule ends. The length of a microtubule remained approximately constant, and only subunits very close to the end would exchange. Then in the late 1970s and

early 1980s, evidence accumulated for a more dynamic activity termed *treadmilling*. The treadmilling model postulated a constant net loss of subunits from one end and a gain at the other. The length of the microtubule would still remain constant, but all subunits would eventually exchange with the soluble pool. Finally, in 1984 a novel dynamic mechanism termed *dynamic instability* was proposed. This model states that the end of a microtubule is never in equilibrium; most of the time the microtubule is in a state of constant growth, but these episodes of elongation are interspersed with episodes of shortening. Since 1984, the phenomenon of dynamic instability has been convincingly established both in vitro and in vivo, and the model can explain the experimental data previously attributed to equilibrium exchange and treadmilling.

Figure 1 illustrates the phenomenon of dynamic instability using quantitative data from the study of Walker et al (75). At or near steady state, each microtubule end is in one of two phases. Most of the microtubules are in the elongation phase and continue growing at a slow, steady rate. A small fraction of the microtubules are in the rapid shortening phase, depolymerizing at a rate about 2–10 times faster than the elongation. A microtubule in either phase remains there for an extended period of time, but occasional transitions occur abruptly and stochastically. The transition from elongation to rapid shortening is termed *catastrophe*, and the reverse transition is termed *rescue*.

Mitchison & Kirschner (47, 48) first inferred dynamic instability from the measurements of fixed microtubules. The phenomenon was then vividly confirmed through direct visualization of single microtubules in vitro by Horio & Hotani (36) using dark field light microscopy and by Walker et al (75) using video-enhanced, differential interference contrast (DIC) light microscopy. This last paper and the related study of O'Brien et al (55) provide comprehensive data on the kinetics of growth rates and transition frequencies in vitro. Finally, Cassimeris et al (19) obtained similar kinetic data for interphase microtubules in living cells. Table 1 summarizes data from these studies.

What is the biological function of dynamic instability? Obviously it produces a constant redistribution of the cytoskeleton, but for what purpose? Perhaps most important is the recycling of microtubules that have missed their target. Specifically, in the mitotic apparatus, microtubules growing from the centrosome attach chromosomes and move them, but only if they spear the kinetochore (33). Those microtubules that miss the kinetochore need to be dismantled and recycled for another shot. Dynamic instability provides the mechanism. Similar redistribution of the cytoskeleton and recycling of microtubules are probably important in interphase, although the targets are not known. Schultze & Kirschner (65)

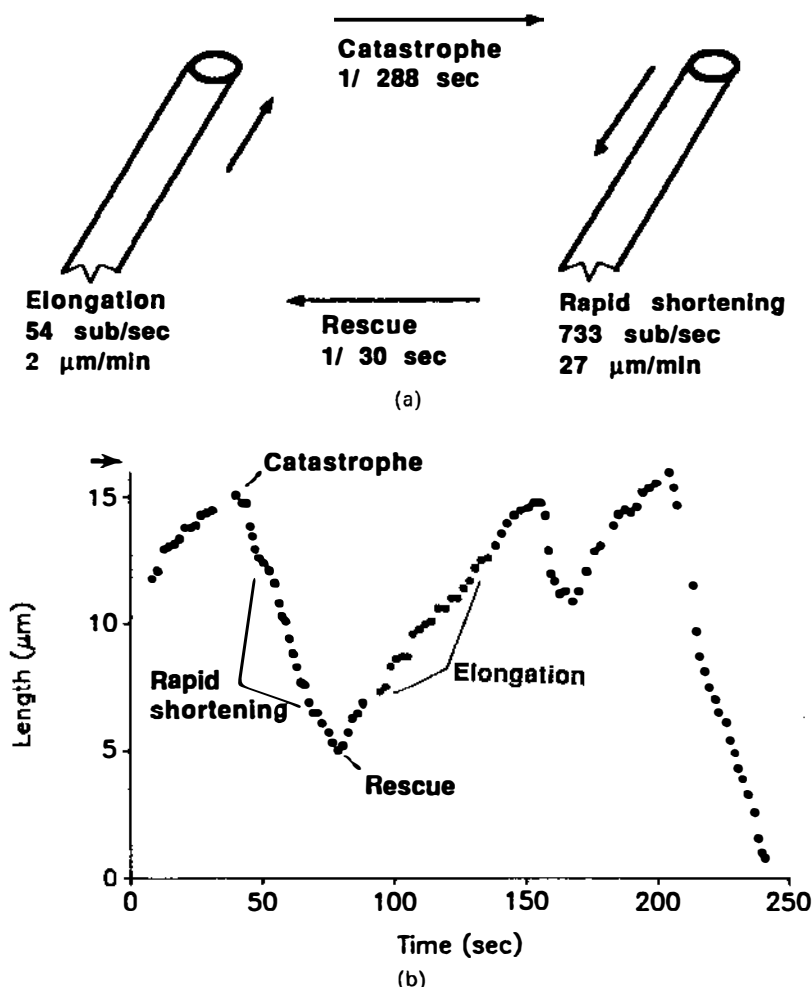


Figure 1 (a) The two phases, elongation and rapid shortening, and the two transitions, catastrophe and rescue, of dynamic instability. The numerical values are rates observed in vitro for the plus end (75) at $11 \mu\text{M}$ free tubulin, the estimated steady-state concentration. (b) Life history of a microtubule undergoing dynamic instability inside a living cell. The rates of growth and rapid shortening are constant for extended periods, and the transitions are abrupt and stochastic. From Cassimeris et al (19).

found that about 20% of the interphase microtubules are stabilized; perhaps these have hit a target in the cell cortex. The remaining dynamic microtubules are still seeking targets and being recycled.

Several excellent reviews have discussed the biological functions of

Table 1 The excursions of dynamic instability^a

	Elongation			Rapid shortening		
	Rate ($\mu\text{m}/\text{min}$)	Time of excursion (min)	Length of excursion (μm)	Rate ($\mu\text{m}/\text{min}$)	Time of excursion (min)	Length of excursion (μm)
Plus end, 11 μM tub, 1 mM Mg	2.0	4.7	9.5	27	0.5	13.5
Minus end, 11 μM tub, 1 mM Mg	0.85	8.3	7.0	34	0.11	3.7
Plus end, 15 μM tub, 1 mM Mg	3.3	14	46	27	0.5	13.5
Minus end, 15 μM tub, 1 mM Mg	1.5	7	25	34	0.08	2.7
Plus end, 15 μM tub, 6 mM Mg	4.2	6.7	28	80	2	160
Minus end, 15 μM tub, 6 mM Mg	2.0	6.7	13.4	170	0.3	51
Plus end, in vivo	7.2	1.2	8.6	17.3	0.37	6.3

^aValues are given for the average growth and disassembly rates, and the average time spent in each phase, as observed by DIC microscopy of single microtubules. Data are from refs. 19, 55, and 75.

dynamic instability (20, 40, 50). Carlier (14, 15) has reviewed the role of nucleotide hydrolysis in the dynamics of microtubules and actin and discusses model mechanisms. The review of Bayley (4) focuses on his “lateral cap” model, a variant of the stochastic dissociation models (75) (see Figure 3c, below). Hill (35) provides an extensive mathematical development of assembly models, including applications to dynamic instability. Salmon et al (61) reviewed the technology of video-enhanced DIC microscopy, which has been the most powerful tool for studying dynamic instability.

The present review begins by relating the modern concept of dynamic instability to the earlier concepts of equilibrium assembly and treadmilling. We then present the concept of a GTP cap as the mechanism governing the phases of dynamic instability, and briefly discuss the evolution of GTP-cap models. In spite of these many advances, a definitive and compelling model is not yet in hand. We therefore conclude by reviewing in detail three avenues of experimental work that appear most promising for unraveling the molecular mechanisms of dynamic instability: (a) kinetics of growth, shortening, and transitions as observed using video microscopy of single microtubules; (b) kinetics of GTP hydrolysis; and (c) the study of GTP analogs.

FROM EQUILIBRIUM ASSEMBLY, THROUGH TREADMILLING, TO DYNAMIC INSTABILITY

The earliest concept of microtubule dynamics was based on the classic theory of condensation polymerization of cooperative helical polymers (57) and upon experimental data such as that shown in Figure 2. When a solution of tubulin subunits is raised from 4°C to 37°C, the subunits spontaneously assemble to form microtubules. After a few minutes, the assembly reaches a stable plateau, as measured by turbidity. The microtubules appear to be in equilibrium with a pool of soluble tubulin subunits; this equilibrium concentration of free subunits has been referred to for years as the “critical concentration” (C_C) (57). We use C_C to refer to an equilibrium polymerization, and the term C_{ss} to designate the steady-state concentration of soluble subunits in the more dynamic nonequilibrium reactions.

If the microtubule were assembled through the reversible association of subunits, this equilibrium polymer would display very little dynamic activity. If a solution of polymers in equilibrium with subunits were diluted, the concentration of soluble subunits would initially fall below C_C , and polymers would then disassemble until the concentration of soluble subunits rose again to C_C . If extra soluble subunits were added, raising the

When the system is at equilibrium, the amount of products and the reactants have not to be necessarily equal. There can be a higher amount of reactants than the products or vice versa. The only requirement in an equilibrium equation is to maintain a constant amount from both over time.

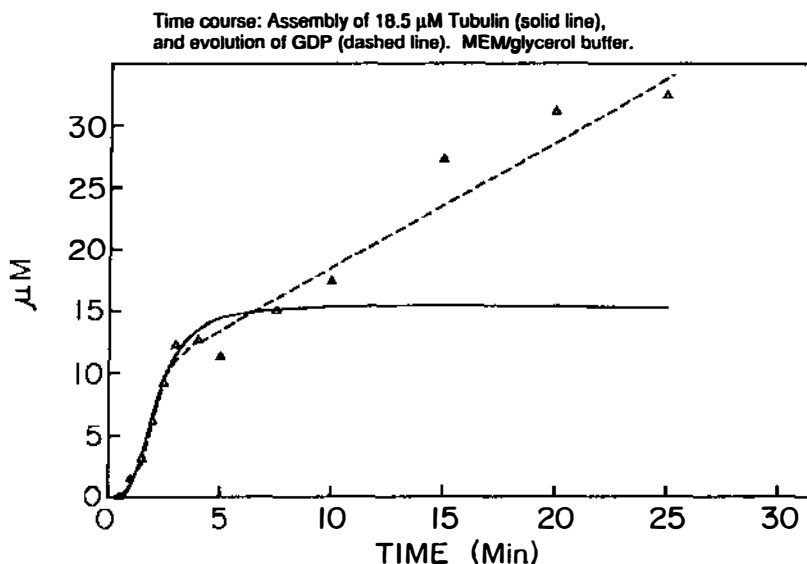


Figure 2 In vitro microtubule assembly followed by turbidity, and simultaneously assayed for GTP hydrolysis. At time 0, the temperature was raised to 37°C to initiate assembly. The turbidity rises to a plateau (steady state) in 4 min. GTP hydrolysis closely follows turbidity during the net assembly phase, and continues at a constant rate at steady state. From O'Brien et al (56).

concentration above C_C , they would assemble to make more polymer, until the subunit concentration fell to C_C . At equilibrium, subunits would constantly exchange on and off the ends of microtubules, but these processes would balance each other. The stochastic nature of the association-dissociation reaction would result in a slow net incorporation of new subunits at each end [proportional to the square root of elapsed time (11)] but these would never extend more than a few hundred subunits into the core of the microtubule. The concept of treadmilling provided for a much more dynamic activity. The theoretical basis of treadmilling is the polar structure of the microtubule; the opposite ends are designated plus and minus. Wegner (77) reasoned that if assembly of a polar polymer were accompanied by an irreversible chemical step like nucleotide hydrolysis, the C_C could be different at the plus and minus ends. The steady-state concentration of free subunits, C_{SS} , would lie between the C_C s of each end. At the minus end of the microtubule, with C_C below C_{SS} , assembly would predominate, and the net gain of subunits at the minus end would be balanced by a net loss at the plus end. Thus, new subunits would be

incorporated one by one at the minus end, treadmill through the microtubule, and eventually exit at the plus end. Experimental data for continued incorporation of subunits at steady state supported this model (42).

We now recognize that the gross effect of treadmilling can be produced by dynamic instability (26, 75). The major new feature of dynamic instability is that, instead of an equilibrium exchange of one subunit at a time, dynamic instability postulates large excursions involving loss or gain of thousands of subunits in each excursion. For example, one may note from the data in Figure 1a that the average length gained during elongation, ($2 \mu\text{m}/\text{min} \times 4.8 \text{ min} = 9.6 \mu\text{m}$), is less than that lost during rapid shortening, ($27 \mu\text{m}/\text{min} \times 0.5 \text{ min} = 13.5 \mu\text{m}$). The plus end, over time, undergoes a net shortening, which is balanced at the minus end, where the excursions of elongation exceed those of shortening. [The specialist should also note the fundamental difference in molecular mechanism. In the Wegner treadmilling model, GTP must be hydrolyzed immediately, leaving GDP-tubulin at the end of the microtubule, while models for dynamic instability postulate a GTP cap at the microtubule end. With this cap, the subunit-by-subunit treadmilling mechanism cannot operate. Thus if treadmilling does occur *in vivo* (49), the mechanism probably involves unbalanced excursions of dynamic instability.] Mitchison & Kirschner (47, 48) discovered the phenomenon of dynamic instability as they tested a simple microtubule assembly system for the predictions of equilibrium assembly. They had devised a preparation of centrosomes that would nucleate microtubules. The centrosomes were placed in a concentrated tubulin solution and allowed to grow clusters of microtubules about $8 \mu\text{m}$ long. The microtubule clusters were then diluted into three different concentrations of soluble tubulin, and the lengths of the microtubules were measured after different time intervals. The classical theory predicted that the microtubules should grow longer when the tubulin concentration was above C_{ss} (which Mitchison & Kirschner assumed was near the C_C of the extended plus ends), and this was what they observed. The classical theory also predicted that microtubules should neither grow nor shrink when the tubulin concentration was approximately equal to C_{ss} , but they found that the microtubules continued to grow. Even more surprising was the experiment diluting the free tubulin below C_{ss} . Whereas the classical theory predicted that all microtubules should shrink at a uniform rate, Mitchison & Kirschner observed that some microtubules seemed to disappear very quickly, but those remaining continued to grow. Thus, the plus end of the microtubule extending from the centrosome can apparently exist in two states with very different C_C s, one favoring elongation and the other favoring shortening. What is the mechanism for maintaining these very different states, and for switching between them?

THE GTP CAP

Origins and Refinements of GTP-Cap Models

The energy source that powers the constant activity of dynamic instability is almost certainly the hydrolysis of GTP. The tubulin dimer binds two molecules of guanine nucleotide. One of these is always GTP, is not exchangeable with nucleotides in solution, and is not hydrolyzed during assembly. This N-site GTP is not discussed here. The second nucleotide binding site (E site) can exchangeably bind either GTP or GDP (or other nucleotides discussed below) from solution. Microtubule assembly under most conditions requires that the E-site nucleotide be GTP, and during assembly the E-site GTP is hydrolyzed to GDP. The E-site GDP is non-exchangeable while the tubulin remains in the polymer. When the microtubule depolymerizes, the soluble tubulin can exchange the GDP for GTP and be reactivated for another cycle of assembly.

How could the energy of GTP hydrolysis be used to generate dynamic instability? The most attractive hypothesis involves some form of GTP cap. If GTP hydrolysis occurs some time after a subunit enters the microtubule lattice, the core of the microtubule will be GDP tubulin, and a small cap of GTP tubulin subunits will be maintained at the ends. The general form of this model postulates that the GTP cap is stable, favoring polymerization, while the GDP core is very labile, favoring rapid disassembly. Thus catastrophe is initiated by loss of the last GTP-tubulin subunit from the cap, and rescue is binding of a GTP subunit to recap a shortening GDP core.

The differential stability of the GTP cap and the GDP core may be related to the two conformations of the tubulin protofilament (46a). When the protofilaments are in the microtubule wall, they are necessarily straight, parallel to each other and to the axis of the microtubule. When a protofilament is freed from the lateral bonding to its neighbors, it tends to adopt a curved conformation, forming a variety of ring polymers (24a, 39a). The curved conformation is enhanced by the presence of GDP at the E site (36a), implying that the straight protofilaments in the GDP core are inherently strained. They are maintained in the straight conformation so long as they are bonded to neighbors on all sides, but when exposed at the end they peel away into the curved conformation. Thus disassembly may be driven by transition to the more favorable curved conformation. GTP tubulin, on the other hand, is relatively stable in the straight conformation (36a) and may form a stable lattice even when exposed at the end of the microtubule. Thus the E-site GTP is proposed to be an allosteric effector that favors the straight conformation, forming a stable cap (46a). A recent paper of Mandelkow et al (41a) provides striking visual evidence of curved

protofilaments peeling away from disassembling microtubules, confirming the earlier negative stain observations (24a, 39a) on a time scale that is rapid and related to the phases of dynamic instability.

The size of the GTP cap is postulated, in most models, to fluctuate through stochastic processes. The actual size at any moment is important in determining the probability of catastrophe. Regardless of the specific model, the size of the GTP cap must be governed by the balance of three reactions: (a) the polymerization of a new GTP-tubulin subunit increases the cap size; (b) dissociation of a GTP-tubulin from the end of the microtubule decreases the cap size; and (c) hydrolysis of GTP also decreases the cap size. In most models, catastrophe occurs only when the last GTP subunit is lost. Any size cap, even a single terminal GTP-tubulin subunit, can stabilize the microtubule and keep it in the elongation phase. However, as the cap grows larger than a single subunit, the probability of achieving catastrophe decreases.

Several models for the GTP cap have been proposed, and these can be grouped into three categories (Figure 3). The ultimate test for these models is the extensive body of data from video microscopy of single microtubules. The most important data, and the most difficult for models to match, are the frequencies of catastrophe over the extended range of free tubulin concentration. Catastrophe frequency decreased only a factor of three as the free tubulin concentration was raised from 7 to 15 μM (75). In contrast, the GTP-cap models all predict a 10- to 100-fold change in catastrophe over this concentration range.

A common feature of the models generates this very steep concentration dependence: the size of the cap, and the probability of its loss, are a balance of stochastic addition and removal of GTP subunits from the cap. However, although the models fail to fit the full range of data, they can all match the frequency of catastrophe at a single concentration. This matching has led to one important conclusion, that parameters must be chosen so that the average cap is very small, no more than about eight subunits (5, 71, 75).

Probing the GTP Cap by Cutting and Diluting Microtubules

Because no specific GTP-cap model can match all the experimental data, one might question the whole idea of a cap as the regulator of dynamic instability. Perhaps catastrophe is generated, not at the end, but by some structural transformation within the microtubule. Two simple tests support the general concept that regulation is at the end, and therefore involves some kind of cap.

A first test involved cutting the microtubule in the middle. According to the model, the newly exposed ends should both be uncapped, and they

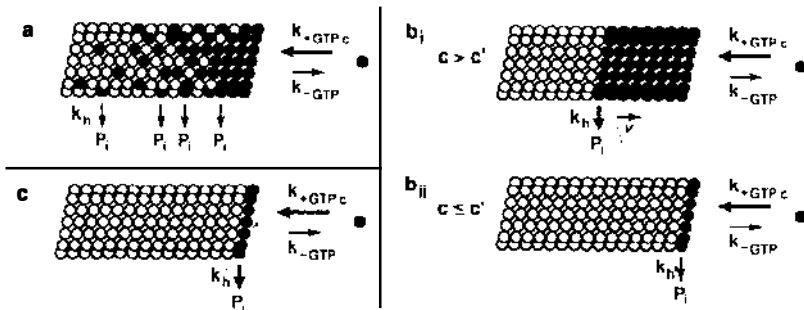


Figure 3 Models for the GTP cap (from 68). Tubulin-GTP subunits are represented by solid black circles, GDP-tubulin by open circles. The models differ in the site and mechanism of GTP hydrolysis. (a) Uncoupled stochastic hydrolysis. This was the original basis for the concept of a GTP cap (12, 47). GTP hydrolysis occurs randomly as a first-order reaction following incorporation of the subunit. During the initial assembly at high subunit concentration, virtually the entire polymer is GTP-tubulin. Later, the core is primarily GDP-tubulin, and the ends are primarily GTP-tubulin. (b) Uncoupled vectorial hydrolysis (8, 16, 46). GTP hydrolysis is enhanced by contact with a GDP-subunit, and therefore occurs almost exclusively at the interface of the GTP cap and the GDP core; moreover, this hydrolysis occurs at a constant velocity. There are two very different regimes. At high tubulin concentrations, the assembly of new subunits exceeds the rate of hydrolysis, and a substantial cap of GTP-tubulin builds up (b_i). At lower tubulin concentrations, where the assembly is slower than the hydrolysis rate, the simplest hypothesis is that the cap disappears, but then the model predicts immediate catastrophe, contrary to observation. An alternative proposal is that hydrolysis slows when only a single ring of subunits is present (b_{ii}); in this case the model reduces to that of panel c. (c) Stochastic dissociation, coupled hydrolysis (75). In this model, the cap cannot exceed a maximum size, say 13 GTP-subunits. The addition of a fourteenth GTP subunit forces the hydrolysis of the deepest one. The actual size of the cap can fluctuate from 13 to 0 subunits, depending on the stochastic balance of association and dissociation of GTP subunits. The lateral cap model (3–5) is closely related but has additional growing points and dissociation parameters.

should immediately begin rapid shortening. Walker et al (74) used a UV microbeam to cut single microtubules in vitro, observing them with DIC optics a few seconds after cutting. The observations were definitive but partly surprising. The newly exposed plus ends all began rapid shortening immediately after cutting, exactly as predicted. However, the newly exposed minus ends never shortened; they continued elongation. These investigators discussed several explanations (74), but the one we favor is that the cutting leaves the minus end in a state that is rapidly rescued. Experiments at higher Mg concentration support this speculation. At 6 mM Mg, we observed that minus-end rescue was reduced by a factor of four (55), and R. A. Walker (73a) found that a substantial fraction of cut

minus ends began rapid shortening when the cutting was done in higher Mg concentrations.

Nicklas et al (52) observed essentially the same behavior, disassembly of plus ends but stable minus ends, in an *in vivo* cutting experiment. They used a microneedle to mechanically sever the microtubules of the mitotic spindle. The exposed plus ends showed substantial depolymerization, but the minus ends were remarkably stable. Cutting with a laser *in vivo* has given similar results (see 52, 69). Overall, we interpret the cutting experiments as consistent with the concept of a labile microtubule core stabilized by caps at the ends, with the severed minus end perhaps rapidly rescued.

A second test of the cap was to dilute the microtubules. If the stabilizing GTP subunits can be lost by dissociation, as predicted in most models, dilution should lead to loss of the GTP cap. Two dilution studies have now demonstrated that microtubules undergo catastrophe very quickly following dilution. Voter et al (72) concluded that all microtubules lost their caps, at both ends, in less than one second following a 15-fold dilution from steady state (to 0.5 μM free tubulin). Walker et al (76) could not achieve as rapid nor complete dilution (using a flow cell to observe single microtubules), but observed catastrophe within about 5 seconds at $\sim 3.3 \mu\text{M}$ free tubulin. Extrapolation of their data to higher dilutions was consistent with the $< 1\text{-s}$ lifetime deduced by Voter et al. Based on this lifetime, the size of the cap was estimated to be no more than 40 and 20 subunits at the plus and minus ends, respectively (72). The data are, of course, consistent with a smaller cap of 1–13 subunits.

Questions for Future Analysis and Models of the GTP Cap

We conclude this section by raising three questions not readily answered by present GTP-cap models. We hope that questions like these might lead to new insights, and eventually to a more compelling model for the mechanism of dynamic instability.

The first question is: why is the frequency of catastrophe approximately the same at the plus and minus ends? In principle, the assembly dynamics could be completely different at the two ends. The polarity problem is clearest at the molecular level: the β -tubulin subunit containing the E-site GTP (41) is exposed at one end, but at the other end it is buried in contact with an α subunit. This arrangement poses a vexing problem for model builders because different rules for coupled GTP hydrolysis generally have to be proposed for the two ends (43). We should also consider the question in terms of biology. If this type of model building is approximately correct, and different mechanisms for GTP hydrolysis indeed apply to the two ends, we must conclude that evolution has generated these separate mechanisms and finely tuned them to produce similar frequencies of catas-

trophe. This seems most unlikely because the minus ends are embedded in centrosomes and probably do not even exhibit dynamic instability *in vivo*. A mechanism that completely ignored the polarity of the lattice and necessarily produced identical frequencies of catastrophe at each end would be much more satisfying. Present models cannot provide such a mechanism.

The second question is: how can we explain the uncoupling of catastrophe from elongation rate, which is apparent in some experiments? In any model with a variable cap size, the cap should grow larger as the elongation rate increases, and one would expect a corresponding decrease in frequency of catastrophe. Indeed, in the original experiments of Walker et al (75), an increase in free tubulin was accompanied by an increase in elongation rate and a decrease in catastrophe. However, at least three observations indicate that the decrease in catastrophe is not directly coupled to elongation: (a) as mentioned above, the plus and minus ends have very different elongation rates, yet essentially the same catastrophe rates; (b) at 6.0 mM Mg, the elongation rates are twofold higher than at 0.5 mM Mg, but the catastrophe frequency did not change (55); (c) in recent experiments (54), we have found that Ca in the range of 1 mM greatly increased the frequency of catastrophe, yet did not affect the growth rate of microtubules that were still elongating. The Mg and Ca experiments argue convincingly against any model in which the cap size is proportional to elongation rate; a satisfactory model should explain how catastrophe and elongation are uncoupled and can be independently modulated.

The third question is: how does dynamic instability apply to an open microtubule sheet, comprising only a few protofilaments? Certainly these sheets, which are intermediates in the nucleation of microtubules and often persist at steady state (25, 67), must exist in a stabilized form. Parsimony would suggest that they are stabilized by the same mechanisms as the intact microtubule. Previously proposed models for a GTP cap have all been drawn for an intact helical lattice, which greatly simplifies several problems. To highlight these problems, the interested reader should try to draw a GTP cap that will stabilize a sheet of, say, four protofilaments, yet still permit elongation and lateral growth. Constructing a GTP cap for a sheet of protofilaments has yet to be considered, but may generate important new insights.

The molecular mechanism by which GTP hydrolysis produces and regulates dynamic instability is the holy grail in this field. GTP hydrolysis must certainly play the key role, and thinking has so far focused on GTP-cap models. However, none of the specific models fits all the experimental data. The basic concept of the GTP cap may be correct, but we simply have not found the correct molecular model. Alternatively, a simple cap

of GTP-tubulin subunits may be fundamentally wrong. New insights are needed, and will perhaps be stimulated by new experimental approaches. The following sections review three experimental areas that may hold the key for necessary new approaches and new molecular insights.

DYNAMIC INSTABILITY IN VITRO—THE FOUR REACTIONS OBSERVED WITH VIDEO MICROSCOPY

The following descriptions and conclusions are drawn primarily from the studies of Walker et al (75), in which the behavior of single microtubules was observed over a range of concentrations of free tubulin, and O'Brien et al (55), in which both tubulin and magnesium were varied. Table 1 gives numerical values for many of the deduced growth rates. Note that to convert $\mu\text{m}/\text{min}$ to subunits per second, we calculate 1625 subunits per μm .

In the elongation phase, microtubules grow at a constant rate that is directly proportional to the concentration of free subunits. A remarkable feature of the data is the negative y -intercept of the line plotting growth rate vs tubulin concentration. This observation indicates reversible association of GTP tubulin onto the growing end, where the size of the intercept gives the dissociation rate. Growth at the minus end is about one-third that at the plus end, and growth at both ends increases by about 50% as Mg is increased from 0.25 mM to 6 mM. In all cases, elongation appears to be a second-order reaction, the reversible association of GTP tubulin subunits to the growing end.

The rate of rapid shortening was found to be independent of the concentration of soluble tubulin and therefore appeared to involve a first-order dissociation of subunits from the microtubule end. The dissociating subunits must be GDP tubulin, because measurements of bulk polymer show only GDP on the E site in microtubule polymer. Walker et al (75) reported shortening at about 30 $\mu\text{m}/\text{min}$ at both ends, but the study of O'Brien et al (55) revealed a much more complex situation. Although the shortening rate was constant along an extended length of most microtubules, the rate varied significantly among the microtubules in a population, and sometimes changed abruptly along a single microtubule. The wide range of shortening rates was most dramatic at higher Mg concentrations, where minus-end shortening varied from 60 to 250 $\mu\text{m}/\text{min}$. A possible explanation for the different shortening rates is that in vitro assembly produces microtubules with 13, 14, and 15 protofilaments, the number sometimes changing within a single microtubule (58, 73). Microtubules with 14 or 15 protofilaments might be significantly strained in the lattice, leading to more rapid dissociation of subunits. Alternatively, some

of the shortening polymers might be open sheets or hooked microtubules (67), with very different disassembly rates.

Catastrophe occurs infrequently and apparently at random, but the transition is always abrupt at the resolution of the light microscope (i.e. an imminent transition is not preceded by a slowing growth). The frequency of catastrophe, the reciprocal of the average time of the elongation phase, decreases in an apparently linear fashion by about a factor of three over the twofold range of free tubulin concentration that could be observed experimentally. The recent dilution studies (72, 76) now extend the range to lower concentrations. Remarkably, the frequency of catastrophe at the minus end is approximately the same as at the plus end at all tubulin concentrations (75, 76) and at 6 and 1 mM Mg (55).

Rescue, like catastrophe, is infrequent, abrupt, and apparently stochastic. The experimental data are probably weakest on the rescue reaction. Although Walker et al (75) concluded that rescue was proportional to tubulin concentration, the data are very scattered, and O'Brien et al (55) could not demonstrate any significant dependence of rescue on tubulin concentration. The mechanism of rescue is therefore obscure. If it does depend on tubulin concentration it should involve a second-order association of a GTP tubulin onto the GDP core, recapping the disassembling microtubule. On the other hand, if rescue does not depend on free tubulin, it might be caused by something within the microtubule lattice, such as a defect that halted disassembly long enough for recapping, or perhaps a buried GTP-tubulin subunit. Rescue must be related in an important way to the GTP cap, but more experimental data are needed on this difficult reaction.

KINETICS OF GTP HYDROLYSIS AND A WORD ABOUT G-PROTEINS

The best kinetic data on GTP hydrolysis have been obtained by nucleotide analysis of assembly reactions at small time intervals after assembly initiation. Assembly is initiated by raising the temperature to 37°C; polymer assembly is estimated by turbidity; and samples are quenched into acid at different time points for subsequent analysis of hydrolyzed GTP. The time resolution for these pipet-determined points is 2–10 s. A typical assembly (Figure 2) shows two phases. During the net assembly phase, the first 3–4 min, GTP hydrolysis is approximately stoichiometric—one GTP is hydrolyzed per tubulin dimer incorporated into polymer—and apparently occurs simultaneously with or shortly following the incorporation of each subunit into the polymer. When steady state is reached, GTP hydrolysis continues at a constant and substantial rate. This steady-state GTP

hydrolysis presumably reflects the continued assembly and disassembly of microtubules undergoing dynamic instability.

How closely is hydrolysis coupled to assembly? The early study of Carlier & Pantaloni (12) was one of the most important inspirations for the GTP-cap model because it concluded that GTP hydrolysis occurred several minutes after incorporation of the subunit. This observation was the basis for the uncoupled stochastic hydrolysis model (Figure 3a). However, the reported lag would generate a very large cap, which would be impossible to lose by stochastic processes to yield the observed frequency of catastrophe (56, 75). More recent studies by O'Brien et al (56) and Carlier et al (16) concluded that the magnitude of the lag was greatly overestimated in the earlier study. At tubulin concentrations near steady state, both studies agree that hydrolysis is closely coupled to assembly and no lag can be detected at the 2- to 10-s time resolution of the pipetting assay. Some controversy remains at higher tubulin concentrations. Carlier et al (16) reported a substantial build-up of a GTP cap at higher tubulin concentrations, and this was the basis for their second model [vectorial hydrolysis, Figure 3b; see also the paper of Caplow (46)]. In contrast, O'Brien et al (56) found that GTP hydrolysis was closely coupled to assembly at all tubulin concentrations, even up to 74 μM ; i.e. there was no evidence for a buildup of a measurable GTP cap under any experimental condition.

Two recent studies used rapid and sensitive filter binding assays to look for transient incorporation of GTP into microtubules (46, 68). Melki et al (46) reported that, following incorporation of the subunit and hydrolysis of GTP, the Pi was released only very slowly from the microtubule, with a half-time of 50 s. They presented this result as consistent with their model of a GDP-Pi cap (however, one again encounters the problem that a 50-s lag would lead to very large caps, which are very difficult to lose through stochastic processes). The independent study of Stewart et al (68), using a similar filter binding assay, arrived at the opposite conclusion. They found no evidence for any transiently bound Pi, either in GTP or as a bound GDP-Pi intermediate. Technical differences in the procedures used by the two groups might explain the contradictory conclusions, and additional experimental work is needed to resolve this important controversy.

David-Pfeuty et al (23) found that if the microtubules were centrifuged out of a steady-state assembly mixture, the supernatant, comprising soluble tubulin at C_{ss} , had no remaining GTPase activity. Caplow & Shanks (10) used chemically stabilized microtubule seeds to study GTP hydrolysis at concentrations of free subunits below C_{ss} . They found that the GTPase rate depended linearly on the concentration of both microtubule seeds and of free subunits. Thus, the major hydrolysis pathway appears to require

interaction of soluble GTP-tubulin with a microtubule end. An important related study is of the very prominent GTPase of tubulin-colchicine (1, 2, 62, 63). Although the tubulin-colchicine assembles into large aggregates of ring polymers, the GTP hydrolysis is apparently not coupled to this assembly. However, a separate study (34) showed that hydrolysis depends on the concentration of tubulin-colchicine. The data suggested that association with another subunit (perhaps a precursor to the larger aggregates) is required for hydrolysis. The prevailing idea from many studies is that soluble tubulin subunits do not hydrolyze GTP; hydrolysis occurs only after the subunit is incorporated into a polymer (or perhaps a dimer for tubulin-colchicine).

Analysis of steady-state GTP hydrolysis is a potentially important, but so far under-utilized, tool in dynamic instability research. Hydrolysis of GTP should equal the total rate of addition of new subunits to microtubules in the elongation phase. Buffer conditions such as 3 M glycerol or 1 M sodium glutamate, which stabilize microtubules, are generally thought to inhibit dynamic instability. However, the steady-state GTPase rate is at least as high in these buffers (29, 56) as in those in which vigorous dynamic instability is demonstrated. Even more remarkable are the effects of the drug taxol, which greatly stabilizes microtubules while only slightly reducing the steady-state GTP hydrolysis (13, 28). This result suggests that dynamic instability continues in these stabilizing buffers, and that the rate of elongation (which is coupled to GTP hydrolysis in the favored GTP-cap models) is not diminished. Because elongation must be balanced by shortening at steady state, these buffers should primarily affect catastrophe and rescue, perhaps greatly reducing the size of the excursions. Direct video microscopy of single microtubules in stabilization buffers and taxol, coupled with measurements of bulk steady-state GTP hydrolysis, may settle this speculation or may reveal new features of dynamic instability.

Many proteins besides tubulin bind and hydrolyze GTP. A large family of GTPase proteins has been identified, all of which share a consensus sequence related to GTP binding. These include G-proteins, elongation factor-Tu, and proteins involved in protein secretion and trafficking (7, 24). In all of these systems, GTP hydrolysis appears to provide a clock mechanism, activating and deactivating the protein after an extended but limited time interval. GTP hydrolysis might also be thought of as a clock in tubulin, determining the time spent in the elongation phase. Surprisingly, tubulin does not share the GTP binding consensus sequence of these other proteins (24). Therefore, tubulin is apparently not related to these other proteins and has devised its own mechanisms for GTP binding and hydrolysis. It is not surprising that the effects of GTP analogs (discussed below) may be quite different in tubulin and G-proteins. Nevertheless, the

body of research on the large family of GTPases is very large (7). Many of the techniques employed in these studies could be applied to tubulin, and some of the mechanisms may be related.

One mechanistic feature seems to be shared by almost all members of the GTPase family, as well as by tubulin. The isolated, soluble GTPases hydrolyze GTP at very slow rates, but hydrolysis is greatly accelerated when the protein interacts with a separate GTPase activating protein (GAP) (7). As discussed above, soluble tubulin also has a minimal GTPase activity; GTP is hydrolyzed only after the tubulin is incorporated into polymer. Thus, tubulin may be considered to act as its own GAP, accelerating GTP hydrolysis when specific protein-protein interactions are established in the microtubule.

MICROTUBULE ASSEMBLY WITH GTP ANALOGS

Because microtubule assembly requires GTP-tubulin and the GTP is hydrolyzed in the assembly reaction, there has been considerable interest in nonhydrolyzable GTP analogs. Would these analogs support assembly, and if so would the analog microtubules be more or less stable than the GTP/GDP microtubules? If a microtubule could be assembled from a nonhydrolyzable analog, we might expect it to behave as a giant GTP cap, opening the way to otherwise impossible analyses. We first discuss weakly binding analogs, then analogs that bind more strongly, and finally aluminum and beryllium fluoride.

The most studied analogs are GMPPNP and GMPPCP, in which the oxygen bridging the β and γ phosphates is replaced with nitrogen or carbon. Since the bond lengths and angles of the terminal phosphates are very similar in the analogs and in GTP, the analogs should bind to the E site and be defective only in the hydrolysis step. Several earlier studies (reviewed in 53) gave conflicting results as to whether these analogs would support assembly. We reinvestigated the question using purified tubulin freed of all unbound nucleotide by gel filtration. This tubulin still contained one molecule of GTP bound tightly to each E site, which was sufficient for one round of assembly in the presence or absence of a large excess of analog. However, neither of the analogs would support a second cycle of assembly, and we could obtain no evidence that the analogs bound to the tubulin. The simplest interpretation, that these analogs cannot support microtubule assembly, is, however, not correct. The problem is that the affinity of these analogs for the tubulin E site is orders of magnitude weaker than that of GTP and GDP. Following our first cycle of assembly, the tubulin was left with a stoichiometric amount of E-site GDP, which

could not be competed off even by a large excess of analog. Three recent studies now provide methods for stripping the E-site GTP or GDP and replacing it with the weakly binding analogs (30, 45, 66).

Mejillano et al (45) and Seckler et al (66) developed the alkaline phosphatase technique [first used by Terry & Purich (70)] to digest the E-site GTP or GDP, and demonstrated that GMPPCP and GMPPNP could bind to the vacant E site, could support assembly of microtubules, and was incorporated into the polymer. The latter study carefully documented the rate of digestion by alkaline phosphatase and demonstrated multiple cycles of reversible assembly. Analog microtubules have not been examined yet using video microscopy, but in one study, GMPPCP eliminated the length redistribution characteristic of dynamic instability (44). Alternate procedures for stripping the E-site nucleotide are gel filtration (44) or chromatography on phenyl Sepharose (32), although the latter procedure has not yet been exploited for analog studies. The third recent analog study was that of Hamel & Lin (30), who obtained incorporation and multiple cycles of assembly by shifting to a lower pH [pH 6.1 vs the pH 6.6 used by O'Brien et al (53)]. Finally, note that ATP would also support assembly when incorporated into a vacant E site (30, 45). The ATP was hydrolyzed during the assembly reaction.

GTP γ S and GTP β S, in which one of the nonbridging atoms of the γ or β phosphate are replaced by sulfur, have been important tools in G-protein work (78). Because the bridging oxygen atoms are still in place, hydrolysis can occur, but it is generally much slower than with GTP. Roychowdhury & Gaskin (59; see this paper for references to earlier studies) demonstrated formation of microtubules and sheets in the presence of GTP γ S, provided the Mg concentration was 0.5–1.0 mM. At higher Mg concentrations, the analog induced rings and crystal-like aggregates of rings. Although this study did not demonstrate repeated cycles of assembly, these authors did report that the analog was incorporated into protein polymer, 0.4–0.8 mols of analog per tubulin dimer, and that the GTP γ S was mostly not hydrolyzed. Other modifications of the γ phosphate, such as GTP γ F (51) or bulkier substituents (39), gave analogs that bound strongly to tubulin but inhibited assembly. The GTP β S(A) and -(B) stereoisomers both supported assembly. The A isomer was hydrolyzed, but the B isomer was not (60) and is therefore a good candidate for further study.

Hamel and colleagues (27, 31) have investigated several deoxyguanosine analogs. Deoxy- and dideoxyguanosine (ddGTP) supported microtubule assembly more vigorously than GTP and were hydrolyzed. More remarkable, ddGDP also supported assembly and was incorporated into the microtubule polymer. Apparently, ddGDP displaced the GDP bound to the E site at concentrations above 0.5 mM and is therefore one of the

most strongly binding of the nonhydrolyzable analogs. This analog clearly deserves additional attention.

Another analog that apparently binds strongly enough to displace the E-site GDP, and that supports multiple cycles of assembly, is GMPCPP (53, 64). Although earlier work indicated that this analog was hydrolyzed (64), recent work in our lab (H. P. Erickson & E. T. O'Brien, unpublished data) and by A. Hyman & T. J. Mitchison (personal communication) show that GMPCPP is incorporated stoichiometrically into microtubules and is not hydrolyzed. Hyman et al (38) describe a protocol for synthesizing this analog and reported that microtubules grown in an excess of this analog were extremely stable following dilution at 37°C. Although one might expect the analog microtubules to be more stable than ones grown in GTP, the apparent magnitude of this stability is surprising. First, microtubules with GMPCPP appear to be much more stable than the GTP cap itself, which was estimated to have a C_C of about 3–5 μM and a dissociation rate constant of 44 subunits/s (75). A relatively rapid dissociation rate for the GTP subunits was also deduced in dilution studies (72, 76). Second, the C_C for GMPCPP microtubules was also in the range of 3–5 μM (66); i.e. similar to that proposed for the GTP cap and much higher than that reported for GMPCPP microtubules. Further study of the possible contradictions in C_C and dissociation rates for the analog microtubules may lead to important new insights.

Aluminum and beryllium fluoride activate G-proteins, apparently by creating a kind of GTP analog from bound GDP. When the bound nucleotide is GDP, the G-protein is normally inactivated. However AlF_4^- will bind next to the β phosphate of GDP, mimicking the γ phosphate of GTP and activating the G-protein (6, 21). Three laboratories have now studied the effects of Al, Be, or Pi on microtubule assembly. Humphreys & McDonald (37) reported that AlF_x , in the presence of excess GDP, did not affect assembly. They concluded that the GDP- AlF_x complex did not function as a GTP analog in tubulin. In contrast, Carlier et al (17, 18, 22) reported that both BeF_x and AlF_x could stabilize microtubules against dilution-induced depolymerization. Beryllium bound to microtubules (but not to soluble tubulin) in a 1:1 stoichiometric ratio, and BeF_x stimulated assembly in excess GDP. This assembly was, however, quite slow and probably quite different from the assembly via GTP-tubulin; the BeF_x bound to tubulin may mimic a protein-coordinated free Pi, rather than be a GTP analog. Following this argument, Carlier and colleagues suggested that release of Pi following hydrolysis may be slow, and that a GDP-Pi cap, rather than a GTP cap, may provide the major stabilization of the microtubule end. Caplow et al (9) pursued this hypothesis using DIC light microscopy to study the effect of Pi on the dynamics of single microtubules.

In contrast to the prediction of Carlier et al, they found that 0.167 M Pi had virtually no effect on any of the parameters of dynamic instability; in particular, Pi did not slow rapid shortening, nor did it reduce the frequency of catastrophe. Thus, the direct analysis of single microtubules does not support the idea of a GDP-Pi cap. Nevertheless, the Be and Al fluorides apparently profoundly affect microtubules, and may be important tools in sorting out the complex relations of GTP hydrolysis, Pi release, and microtubule stabilization.

PROSPECTS—THE IMPORTANT QUESTIONS REMAINING

That dynamic instability describes the assembly dynamics of most microtubules is now well established. Many details of the reactions have been determined experimentally, providing important clues to the molecular mechanism. The energy source that powers dynamic instability must be derived from hydrolysis of GTP, and models involving a GTP cap are the most attractive. However the definitive molecular model—which should be based on simple and testable assumptions and should fit all the experimental data—is not yet in hand. The simple idea of a GTP cap may need just a clever modification, or it may need a complete rethinking.

New concepts are likely to come from new experimental approaches. Nonhydrolyzable analogs would seem to offer great potential. A microtubule polymerized completely from analog tubulin should be dynamically inactive and should behave as a pure GTP cap. Even more interesting would be a microtubule doped with a small fraction of analog subunits: would these behave as embedded GTP caps and rescue the microtubule as soon as they are exposed at the end? Or would they reveal a continued dynamics that would force us to reevaluate our ideas of the GTP cap? Also, the kinetics of GTP hydrolysis need to be studied with much higher time resolution, for example, with a quench-flow technology that could provide data on a millisecond time scale. Designing a molecular model is still the ultimate challenge, and in these efforts we should keep in mind the numerous studies of microtubule assembly accumulated over the past two decades. Although the contradictions in present data sometimes seem overwhelming, a simple molecular mechanism certainly exists. The search for this model should provide exciting biochemistry and hopefully a definitive mechanism within the near future.

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